

# Linear Algebra Review

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## Notation: Sets and Graphs

- $\mathbb{A}$ : A set
- $\mathbb{R}$ : Set of real numbers
- $\{0, 1, 2, \dots, n\}$ : Set of all integers between 0 and  $n$
- $[a, b]$ : Real intervals including  $a$  and  $b$
- $(a, b]$ : Interval excluding  $a$  but including  $b$
- $\mathbb{A} \setminus \mathbb{B}$ : Set subtraction

- $a_i$ : Element  $i$  of vector  $\mathbf{a}$ , with indexing starting at 1
- $\mathbf{a}_{-i}$ : All elements of  $\mathbf{a}$  except for element  $i$
- $\mathbf{A}_{i,:}$ : Row  $i$  of matrix  $\mathbf{A}$
- $\mathbf{A}_{:,j}$ : Column  $j$  of matrix  $\mathbf{A}$
- $\mathbf{A}_{i,j,k}$ : Element  $(i, j, k)$  of a 3-D tensor  $\mathbf{A}$
- $\mathbf{A}_{:,:,k}$ : 2-D slice of a 3-D tensor

## Linear Algebra Operations

- $\mathbf{I}_n$ : Identity matrix with  $n$  rows and  $n$  columns
- $\mathbf{A}^T$ : Transpose of matrix  $\mathbf{A}$
- $\mathbf{A}^+$ : Moore-Penrose pseudoinverse of  $\mathbf{A}$
- $\mathbf{A} \odot \mathbf{B}$ : Element-wise (Hadamard) product of  $\mathbf{A}$  and  $\mathbf{B}$
- $\det(\mathbf{A})$ : Determinant of  $\mathbf{A}$
- $\mathbf{x} \in \mathbb{R}^n$ : A vector with  $n$  entries
- $\mathbf{A} \in \mathbb{R}^{m \times n}$ : A matrix with  $m$  rows and  $n$  columns, entries of  $\mathbf{A}$  are real numbers

Study of linear algebra involves following types of mathematical objects

- Scalars- just a single number
- Vectors-it is an array of numbers
- Matrices-a 2-D array of numbers
- Tensors-are useful for representing higher order relations, for example when we need array with more than two axes

- Product of two matrices  $A \in \mathbb{R}^{m \times n}$  and  $B \in \mathbb{R}^{n \times p}$  is the matrix  $C = AB \in \mathbb{R}^{m \times p}$ , where

$$C_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

- Given two vectors  $x, y \in \mathbb{R}^n$ , quantity  $x^T y$  called **inner product** or dot product of vectors, is a real number given by

$$x^T y \in \mathbb{R} = [x_1, x_2, \dots, x_n] \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \sum_{i=1}^n x_i y_i$$

- Note that,  $x^T y = y^T x$
- Given vectors  $x \in \mathbb{R}^m, y \in \mathbb{R}^n$  (not of same size),  $xy^T \in \mathbb{R}^{m \times n}$  is called the **outer product**

$$xy^T \in \mathbb{R}^{m \times n} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} [y_1 \quad y_2 \quad \dots \quad y_n] = \begin{bmatrix} x_1 y_1 & x_1 y_2 & \dots & x_1 y_n \\ x_2 y_1 & x_2 y_2 & \dots & x_2 y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_m y_1 & x_m y_2 & \dots & x_m y_n \end{bmatrix}$$

# Usage of Outer Product

- Let  $\mathbf{1} \in \mathbb{R}^n$ , and  $A \in \mathbb{R}^{m \times n}$  whose columns are all equal to some vector  $x \in \mathbb{R}^m$
- Using outer product, we can represent  $A$  compactly as,

$$A = \begin{bmatrix} x_1 & x_1 & \dots & x_1 \\ x_2 & x_2 & \dots & x_2 \\ \vdots & \vdots & \ddots & \vdots \\ x_m & x_m & \dots & x_m \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix} \begin{bmatrix} 1 & 1 & \dots & 1 \end{bmatrix} = x \mathbf{1}^T$$

## Matrix-Vector Products

Given matrix  $A \in \mathbb{R}^{m \times n}$  and a vector  $x \in \mathbb{R}^n$ , their product  $y = Ax \in \mathbb{R}^m$

$$\bullet y = Ax = \begin{bmatrix} - & a_1^T & - \\ - & a_2^T & - \\ & \vdots & \\ - & a_m^T & - \end{bmatrix} x = \begin{bmatrix} a_1^T x \\ a_2^T x \\ \vdots \\ a_m^T x \end{bmatrix}$$

$$\bullet y = Ax = \begin{bmatrix} | & | & & | \\ a_1 & a_2 & \dots & a_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = [a_1]x_1 + [a_2]x_2 + \dots + [a_n]x_n.$$

$$\bullet y^T = x^T A = x^T \begin{bmatrix} | & | & & | \\ a_1 & a_2 & \dots & a_n \\ | & | & & | \end{bmatrix} = [x^T a_1 \quad x^T a_2 \quad \dots \quad x^T a_n]$$

- Identity matrix  $I \in \mathbb{R}^{n \times n}$ , defined as

$$I_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

- For all  $A \in \mathbb{R}^{m \times n}$ ,  $AI = A = IA$
- Diagonal matrix is a matrix where all non-diagonal elements are 0, and denoted  $D = \text{diag}(d_1, d_2, \dots, d_n)$  with

$$D_{ij} = \begin{cases} d_i & i = j \\ 0 & i \neq j \end{cases}$$

- $I = \text{diag}(1, 1, \dots, 1)$
- $A \in \mathbb{R}^{n \times n}$  is symmetric if  $A = A^T$ , and it is **anti-symmetric** if  $A = -A^T$
- $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$
- $A \in \mathbb{S}^n$  means that  $A$  is a symmetric  $n \times n$  matrix

## Trace

**Trace** of a square matrix  $A \in \mathbb{R}^{n \times n}$ , denoted by  $\text{tr}(A)$  (or  $\text{tr}A$ ) is the sum of diagonal elements in the matrix

- $\text{tr}A = \sum_{i=1}^n A_{ii}$
- For  $A, B, C \in \mathbb{R}^{n \times n}$ , it has following properties
  - $\text{tr}A = \text{tr}A^T$
  - $\text{tr}(A + B) = \text{tr}A + \text{tr}B$
  - $\alpha \in \mathbb{R}, \text{tr}(\alpha A) = \alpha \text{tr}A$
  - For  $A, B$  such that  $AB$  is square  $\text{tr}AB = \text{tr}BA$
  - For  $A, B, C$  such that  $ABC$  is square  $\text{tr}ABC = \text{tr}BCA = \text{tr}CAB$

A **norm** of a vector  $\|x\|$  is a measure of the “length” of the vectors

- Commonly used Euclidean or  $L_2$  norm is defined as

$$\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$$

- Note that  $\|x\|_2^2 = x^T x$
- A norm is any function  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  that satisfies the following properties
  - For all  $x \in \mathbb{R}^n, f(x) \geq 0$  (non-negative)
  - $f(x) = 0$  iff  $x = 0$  (definiteness)
  - For all  $x \in \mathbb{R}^n, \alpha \in \mathbb{R}, f(\alpha x) = |\alpha|f(x)$  (homogeneity)
  - For all  $x, y \in \mathbb{R}^n, f(x + y) \leq f(x) + f(y)$  (triangle inequality)

- $L_1$  and  $L_\infty$  norm is defined as

$$\|x\|_1 = \sum_{i=1}^n |x_i|, \quad \|x\|_\infty = \max_i |x_i|$$

- In general,  $L_p$  norm is defined as

$$\|x\|_p = \left( \sum_{i=1}^n |x_i|^p \right)^{1/p}$$

- Norm of a matrix, such as Frobenius norm is defined as

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n A_{ij}^2} = \sqrt{\text{tr}(A^T A)}$$

# Linear Independence and Rank

- Set of vectors  $\{x_1, x_2, \dots, x_n\} \in \mathbb{R}^m$  is said to be (linearly) independent if no vectors can be presented as a linear combination of remaining vectors
- Conversely, they are called as dependent, i.e  $x_n = \sum_{i=1}^{n-1} \alpha_i x_i$ , for some  $\alpha_i \in \mathbb{R}$
- Column rank of a matrix  $A \in \mathbb{R}^{m \times n}$  is the number of independent columns of  $A$
- Row rank is the largest number of rows of  $A$  that constitute a linearly independent set
- For any matrix  $A \in \mathbb{R}^{m \times n}$  the column rank is equal to the row rank, and collectively called as rank of  $A$ , denoted by  $\text{rank}(A)$
- Basic properties of rank:
  - For  $A \in \mathbb{R}^{m \times n}$ ,  $\text{rank}(A) \leq \min(m, n)$ . If  $\text{rank}(A) = \min(m, n)$ , then  $A$  is said to be full rank
  - For  $A \in \mathbb{R}^{m \times n}$ ,  $\text{rank}(A) = \text{rank}(A^T)$
  - For  $A \in \mathbb{R}^{m \times n}$ ,  $B \in \mathbb{R}^{n \times p}$ ,  $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$
  - For  $A, B \in \mathbb{R}^{m \times n}$ ,  $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$

## Inverse

- Inverse of a square matrix  $A \in \mathbb{R}^{n \times n}$  is denoted by  $A^{-1}$ ,  
 $A^{-1}A = I = AA^{-1}$
- $A$  is invertible iff it is a non-singular i.e  $\det(A) \neq 0$
- $A$  to be invertible, it must be full rank
- Properties of inverse, all assume that  $A, B \in \mathbb{R}^{n \times n}$ 
  - $(A^{-1})^{-1} = A$
  - $(AB)^{-1} = B^{-1}A^{-1}$
  - $(A^{-1})^T = (A^T)^{-1}$

- Two vectors  $x, y \in \mathbb{R}^n$  are orthogonal if  $x^T y = 0$
- A vector  $x \in \mathbb{R}^n$  is normalized if  $\|x\|_2 = 1$
- A square matrix  $U \in \mathbb{R}^{n \times n}$  is orthogonal if all its columns are orthogonal to each other and are normalized (the columns are then referred to as being **orthonormal**)
- From orthogonality and normality,  $U^T U = I = U U^T$
- If  $U$  is not square but its columns are still orthonormal, then  $U^T U = I \neq U U^T$
- Operating on a vector with an orthogonal matrix will not change its Euclidean norm i.e.,  $\|Ux\|_2 = \|x\|_2$ , for any  $x \in \mathbb{R}^n, U \in \mathbb{R}^{n \times n}$

## Range and Nullspace of a Matrix

- **Span** of a set of vectors  $\{x_1, x_2, \dots, x_n\}$  is the set of all vectors that can be expressed as a linear combination of  $\{x_1, x_2, \dots, x_n\}$  i.e.,

$$\text{span}(\{x_1, x_2, \dots, x_n\}) = \left\{ v : v = \sum_{i=1}^n \alpha_i x_i, \quad \alpha_i \in \mathbb{R} \right\}$$

- **Projection** of a vector  $y \in \mathbb{R}^m$  onto span of  $\{x_1, x_2, \dots, x_n\}$  is the vector  $v \in \text{span}(\{x_1, x_2, \dots, x_n\})$ , such that  $v$  is as close to  $y$  as measured by the Euclidean norm  $\|v - y\|_2$
- Projection is denoted as

$$\text{Proj}(y; \{x_1, x_2, \dots, x_n\}) = \underset{v \in \text{span}(\{x_1, x_2, \dots, x_n\})}{\text{argmin}} \|y - v\|_2$$



- **Range** of a matrix  $A \in \mathbb{R}^{m \times n}$ , denoted by  $\mathcal{R}(A)$  is the span of columns of  $A$

$$\mathcal{R}(A) = \{v \in \mathbb{R}^m : v = Ax, x \in \mathbb{R}^n\}$$

- **Nullspace** of a matrix  $A \in \mathbb{R}^{m \times n}$ , denoted as  $\mathcal{N}(A)$  is the set of all vectors that equal 0 when multiplied by  $A$  i.e.,

$$\mathcal{N}(A) = \{x \in \mathbb{R}^n : Ax = 0\}$$

- Vectors in  $\mathcal{R}(A)$  are of size  $m$ , while vectors in  $\mathcal{N}(A)$  are of size  $n$

- Vectors in  $\mathcal{R}(A^T)$  and  $\mathcal{N}(A)$  are both in  $\mathbb{R}^n$  i.e

$$\{w : w = u+v, u \in \mathcal{R}(A^T), v \in \mathcal{N}(A)\} = \mathbb{R}^n, \quad \mathcal{R}(A^T) \cap \mathcal{N}(A) = \{0\}$$

- $\mathcal{R}(A^T)$  and  $\mathcal{N}(A)$  are disjoint subsets that together span the entire space of  $\mathbb{R}^n$
- Sets of this type are called orthogonal complements, and denoted by  $\mathcal{R}(A^T) = \mathcal{N}(A)^\perp$

## Determinant

- Determinant of  $A \in \mathbb{R}^{n \times n}$  is a function  $\det : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$  and is denoted by  $|A|$  or  $\det A$
- $|A| = \sum_{i=1}^n (-1)^{i+j} a_{ij} |A_{\setminus i, \setminus j}|$ , where  $A_{\setminus i, \setminus j}$  is the matrix that results from deleting  $i$ th row and  $j$ th column from  $A$
- Determinant satisfies the following properties
  - Determinant of the identity is 1,  $|I| = 1$
  - Given a matrix  $A \in \mathbb{R}^{n \times n}$ , if we multiply a single row in  $A$  by a scalar  $\alpha \in \mathbb{R}$ , then  $\alpha|A|$
  - If we exchange any two rows of  $A$ , then the determinant of the new matrix is  $-|A|$
- For  $A \in \mathbb{R}^{n \times n}$ ,  $|A| = |A^T|$
- For  $A, B \in \mathbb{R}^{n \times n}$ ,  $|AB| = |A||B|$
- For  $A \in \mathbb{R}^{n \times n}$ ,  $|A| = 0$  iff  $A$  is singular
- For  $A \in \mathbb{R}^{n \times n}$  and  $A$  non-singular,  $|A^{-1}| = 1/|A|$

# Quadratic Forms and Positive Semidefinite Matrices

- Given  $A \in \mathbb{R}^{n \times n}$  and a vector  $x \in \mathbb{R}^n$ , the scalar value  $x^T A x$  is called a **quadratic form**
- $$x^T A x = \sum_{i=1}^n x_i (A x)_i = \sum_{i=1}^n x_i \left( \sum_{j=1}^n A_{ij} x_j \right) = \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j$$
- Some definitions
  - A symmetric matrix  $A \in \mathbb{S}^n$  is **positive definite** (PD) if for all non-zero vectors  $x \in \mathbb{R}^n$ ,  $x^T A x > 0$
  - A symmetric matrix  $A \in \mathbb{S}^n$  is **positive semidefinite** (PSD) if for all vectors  $x \in \mathbb{R}^n$ ,  $x^T A x \geq 0$
  - A symmetric matrix  $A \in \mathbb{S}^n$  is **negative definite** (ND) if for all vectors  $x \in \mathbb{R}^n$ ,  $x^T A x < 0$
  - A symmetric matrix  $A \in \mathbb{S}^n$  is **negative semidefinite** (NSD) if for all vectors  $x \in \mathbb{R}^n$ ,  $x^T A x \leq 0$
  - Finally, a symmetric matrix  $A \in \mathbb{S}^n$  is **indefinite** if it is neither PSD nor NSD.
  - Given  $A \in \mathbb{R}^{n \times n}$ , the matrix  $G = A^T A$  (also known as **Gram matrix**) is always positive semidefinite

## Positive Definite Matrices

**Definition:** A symmetric matrix  $A \in \mathbb{S}^n$  is **positive definite** (PD) if:

$$x^T A x > 0 \quad \text{for all non-zero } x \in \mathbb{R}^n$$

**Interpretation:**

- Geometrically: The quadratic form  $x^T A x$  defines an **ellipsoid**
- All eigenvalues of  $A$  are **strictly positive**
- All leading principal minors are **positive**

## Example: Checking Positive Definiteness

Consider the symmetric matrix:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

**Step 1: Check quadratic form  $x^T Ax > 0$  for all  $x \neq 0$**

Let  $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ :

$$\begin{aligned} x^T Ax &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 2x_1 + x_2 \\ x_1 + 2x_2 \end{bmatrix} \\ &= 2x_1^2 + x_1x_2 + x_1x_2 + 2x_2^2 \\ &= 2x_1^2 + 2x_1x_2 + 2x_2^2 \\ &= 2(x_1^2 + x_1x_2 + x_2^2) \end{aligned}$$

contd...

Complete the square

$$\begin{aligned} x^T Ax &= 2(x_1^2 + x_1x_2 + x_2^2) \\ &= 2\left(x_1^2 + x_1x_2 + \frac{1}{4}x_2^2 + \frac{3}{4}x_2^2\right) \\ &= 2\left[\left(x_1 + \frac{1}{2}x_2\right)^2 + \frac{3}{4}x_2^2\right] \\ &= 2\left(x_1 + \frac{1}{2}x_2\right)^2 + \frac{3}{2}x_2^2 \end{aligned}$$

Since both terms are non-negative and cannot both be zero simultaneously when  $x \neq 0$ :

$$x^T Ax > 0 \quad \text{for all } x \neq 0$$

**A is positive definite.**

## Alternative Check: Eigenvalues

### Verify using eigenvalues

Characteristic equation:

$$\begin{aligned}\det(A - \lambda I) &= \begin{vmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{vmatrix} \\ &= (2 - \lambda)^2 - 1 \\ &= \lambda^2 - 4\lambda + 3 = 0\end{aligned}$$

Eigenvalues:

$$\lambda_{1,2} = \frac{4 \pm \sqrt{16 - 12}}{2} = \frac{4 \pm 2}{2}$$

$$\lambda_1 = 1 > 0, \quad \lambda_2 = 3 > 0$$

**Both eigenvalues positive**  $\Rightarrow A$  is positive definite.

## Alternative Check: Principal Minors

### Verify using leading principal minors

1st leading principal minor:  $\Delta_1 = a_{11} = 2 > 0$

2nd leading principal minor:  $\Delta_2 = \det(A) = \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 4 - 1 = 3 > 0$

**All leading principal minors positive**  $\Rightarrow A$  is positive definite.

### Note

All three methods confirm:  $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$  is positive definite.

A symmetric matrix  $A \in \mathbb{S}^n$  is **positive semidefinite** (PSD) if for all vectors  $x \in \mathbb{R}^n$ ,

$$x^T A x \geq 0.$$

**Notation:**  $A \succeq 0$

- $A$  must be symmetric:  $A = A^T$
- Quadratic form  $x^T A x$  is a scalar
- Inequality must hold for *all*  $x \in \mathbb{R}^n$

## Example: Check if $A$ is PSD

Consider the symmetric matrix:

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$$

**Check:** For any  $x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3$ , compute

$$x^T A x = \begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix} \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

## Example: Computing the Quadratic Form

$$\begin{aligned}x^T Ax &= 2x_1^2 - x_1x_2 - x_1x_2 + 2x_2^2 - x_2x_3 - x_2x_3 + 2x_3^2 \\&= 2x_1^2 - 2x_1x_2 + 2x_2^2 - 2x_2x_3 + 2x_3^2\end{aligned}$$

Rewrite by completing squares:

$$\begin{aligned}x^T Ax &= (x_1^2 - 2x_1x_2 + x_2^2) + (x_1^2 - 2x_1x_2 + x_2^2) \\&\quad + (x_2^2 - 2x_2x_3 + x_3^2) + (x_2^2 - 2x_2x_3 + x_3^2) \\&= (x_1 - x_2)^2 + (x_1 - x_2)^2 + (x_2 - x_3)^2 + (x_2 - x_3)^2 \\&= 2(x_1 - x_2)^2 + 2(x_2 - x_3)^2\end{aligned}$$

## Example: Verification of PSD Property

We have shown:

$$x^T Ax = 2(x_1 - x_2)^2 + 2(x_2 - x_3)^2$$

- Squares are always non-negative:  $(x_1 - x_2)^2 \geq 0$ ,  $(x_2 - x_3)^2 \geq 0$
- Positive coefficients (2) multiply non-negative terms
- Therefore:  $x^T Ax \geq 0$  for *all*  $x \in \mathbb{R}^3$

$A$  is positive semidefinite ( $A \succeq 0$ ) **Note:**  $x^T Ax = 0$  when  $x_1 = x_2 = x_3$

(e.g.,  $x = (1, 1, 1)^T$ ), so  $A$  is *semidefinite*, not positive definite.

## Alternative Check: Eigenvalues

For symmetric matrices, PSD can be verified via eigenvalues:

$$A = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$$

The eigenvalues are:

$$\lambda_1 = 2 - \sqrt{2} \approx 0.586, \quad \lambda_2 = 2, \quad \lambda_3 = 2 + \sqrt{2} \approx 3.414$$

All eigenvalues  $\lambda_i \geq 0 \Rightarrow A$  is PSD. **Theorem:** A symmetric matrix

is PSD if and only if all its eigenvalues are non-negative.

## Eigenvalues and Eigenvectors

- Given  $A \in \mathbb{R}^{n \times n}$ , we say that  $\lambda \in \mathbb{C}$  is an eigenvalue of  $A$  and  $x \in \mathbb{C}^n$  is the corresponding eigenvector if

$$Ax = \lambda x, \quad x \neq 0$$

- To obtain eigenvector corresponding to the eigenvalue  $\lambda_i$ , we solve linear equation  $(\lambda_i I - A)x = 0$
- Properties of eigenvalues and eigenvectors

- $tr A = \sum_{i=1}^n \lambda_i$

- $|A| = \prod_{i=1}^n \lambda_i$

- Rank of  $A$  is equal to the number of non-zero eigenvalues of  $A$
- For  $|A| \neq 0$ ,  $1/\lambda_i$  is an eigenvalue of  $A^{-1}$  with associated eigenvector  $x_i$ , i.e  $A^{-1}x_i = (1/\lambda_i)x_i$
- Eigenvalues of a diagonal matrix  $D = \text{diag}(d_1, \dots, d_n)$  are the diagonal entries  $d_1, \dots, d_n$

- We can write eigenvector equations simultaneously as

$$AX = X\Lambda$$

, where

$$X \in \mathbb{R}^{n \times n} = \begin{bmatrix} | & | & & | \\ x_1 & x_2 & \dots & x_n \\ | & | & & | \end{bmatrix}, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$$

- If the eigenvectors are linearly independent, then  $X$  will be invertible so  $A = X\Lambda X^{-1}$
- A matrix that can be written in this form is called **diagonalizable**

## Matrix Calculus

- Matrix calculus???
- It is the extension of calculus to the vector setting
- Actual calculus is relatively trivial, however, notations are bit different
- We'll particularly focus on
  - Gradient
  - Hessian
  - Gradient and Hessian of quadratic and linear functions
  - Least Squares
  - Gradient of determinant



- Suppose  $f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$ , then the gradient of  $f$  (wrt  $A \in \mathbb{R}^{m \times n}$ ) is a matrix of partial derivatives, defined as

$$\nabla_A f(A) \in \mathbb{R}^{m \times n} = \begin{bmatrix} \frac{\partial f(A)}{\partial A_{11}} & \frac{\partial f(A)}{\partial A_{12}} & \cdots & \frac{\partial f(A)}{\partial A_{1n}} \\ \frac{\partial f(A)}{\partial A_{21}} & \frac{\partial f(A)}{\partial A_{22}} & \cdots & \frac{\partial f(A)}{\partial A_{2n}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f(A)}{\partial A_{m1}} & \frac{\partial f(A)}{\partial A_{m2}} & \cdots & \frac{\partial f(A)}{\partial A_{mn}} \end{bmatrix}$$

- An  $m \times n$  matrix with

$$(\nabla_A f(A))_{ij} = \frac{\partial f(A)}{\partial A_{ij}}$$

- If  $A$  is just a vector  $x \in \mathbb{R}^n$ ,

$$\nabla_x f(x) = \begin{bmatrix} \frac{\partial f(x)}{\partial x_1} \\ \frac{\partial f(x)}{\partial x_2} \\ \vdots \\ \frac{\partial f(x)}{\partial x_n} \end{bmatrix}$$

- Gradient of a function is only defined if the function is real-valued, i.e., it returns a scalar value
- We cannot take the gradient of  $Ax$ ,  $A \in \mathbb{R}^{n \times n}$  wrt  $x$ , as it is vector-valued
- From the properties of partial derivatives
  - $\nabla_x(f(x) + g(x)) = \nabla_x f(x) + \nabla_x g(x)$
  - For  $\alpha \in \mathbb{R}$ ,  $\nabla_x(\alpha f(x)) = \alpha \nabla_x f(x)$
- Gradients are natural extension of partial derivative for multi-variate functions
- Working with gradients can be tricky due to notational reasons

### Example

Let  $f : \mathbb{R}^m \mapsto \mathbb{R}$  be the function defined by

$$f(z) = z^T z$$

then  $\nabla_z f(z) = 2z$ .

- Suppose that  $f : \mathbb{R}^n \mapsto \mathbb{R}$
- Hessian matrix wrt  $x$ , written as  $\nabla_x^2 f(x)$  is given by

$$\nabla_x^2 f(x) \in \mathbb{R}^{n \times n} = \begin{bmatrix} \frac{\partial^2 f(x)}{\partial x_1^2} & \frac{\partial^2 f(x)}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f(x)}{\partial x_2 \partial x_1} & \frac{\partial^2 f(x)}{\partial x_2^2} & \cdots & \frac{\partial^2 f(x)}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f(x)}{\partial x_n \partial x_1} & \frac{\partial^2 f(x)}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_n^2} \end{bmatrix}$$

- $\nabla_x^2 f(x) \in \mathbb{R}^{n \times n}$ , with

$$(\nabla_x^2 f(x))_{ij} = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}$$

- Hessian is always symmetric, since  $\frac{\partial^2 f(x)}{\partial x_i \partial x_j} = \frac{\partial^2 f(x)}{\partial x_j \partial x_i}$
- Gradient is the analogue of first derivative for function
- Hessian as the analogue of second derivative, but with few caveats

## Gradient and Hessian of Quadratic and Linear Function

- For some  $x \in \mathbb{R}^n$ , let  $f(x) = b^T x$  for some  $b \in \mathbb{R}^n$ , then

$$f(x) = \sum_{i=1}^n b_i x_i$$

so

$$\frac{\partial f(x)}{\partial x_k} = \frac{\partial}{\partial x_k} \left( \sum_{i=1}^n b_i x_i \right) = b_k$$

- We can say that  $\nabla_x b^T x = b$

- Consider a quadratic function  $f(x) = x^T Ax$  for  $A \in \mathbb{S}^n$ , which can be written as

$$f(x) = \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j$$

- To take partial derivative, we'll consider term including  $x_k$  and  $x_k^2$  factors separately

$$\begin{aligned} \frac{\partial f(x)}{\partial x_k} &= \frac{\partial}{\partial x_k} \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j \\ &= \frac{\partial}{\partial x_k} \left[ \sum_{i \neq k} \sum_{j \neq k} A_{ij} x_i x_j + \sum_{i \neq k} A_{ik} x_i x_k + \sum_{j \neq k} A_{kj} x_k x_j + A_{kk} x_k^2 \right] \\ &= \sum_{i \neq k} A_{ik} x_i + \sum_{j \neq k} A_{kj} x_j + 2A_{kk} x_k \\ &= \sum_{i=1}^n A_{ik} x_i + \sum_{j=1}^n A_{kj} x_j \\ &= 2 \sum_{i=1}^n A_{ki} x_i \end{aligned}$$

- The  $k$ th entry of  $\nabla_x f(x)$  is the inner product of the  $k$ th row of  $A$  and  $x$
- $\nabla_x x^T Ax = 2Ax$  (think of  $\partial/(\partial x) ax^2 = 2ax$ )
- Hessian of the quadratic function  $f(x) = x^T Ax$

$$\frac{\partial^2 f(x)}{\partial x_k \partial x_l} = \frac{\partial}{\partial x_k} \left[ \frac{\partial f(x)}{\partial x_l} \right] = \frac{\partial}{\partial x_k} \left[ 2 \sum_{i=1}^n A_{li} x_i \right] = 2A_{lk} = 2A_{kl}$$

- So,  $\nabla_x^2 x^T Ax = 2A$
- Recap
  - $\nabla_x b^T x = b$
  - $\nabla_x x^T Ax = 2Ax$  (if  $A$  symmetric)
  - $\nabla_x^2 x^T Ax = 2A$  (if  $A$  symmetric)

# Least Squares

- We are given  $A \in \mathbb{R}^{m \times n}$  (we assume  $A$  is full rank) and vector  $b \in \mathbb{R}^m$  such that  $b \notin \mathcal{R}(A)$
- In this situation we'll not be able to find a vector  $x \in \mathbb{R}^n$ , such that  $Ax = b$
- We want to find a vector  $x$  such that  $Ax$  is as close to  $b$  as possible, as measured by the square of the Euclidean norm  $\|Ax - b\|_2^2$
- Using the fact that  $\|x\|_2^2 = x^T x$ , we have

$$\begin{aligned}\|Ax - b\|_2^2 &= (Ax - b)^T (Ax - b) \\ &= (x^T A^T - b^T)(Ax - b) \\ &= x^T A^T Ax - x^T A^T b - b^T Ax + b^T b \\ &= x^T A^T Ax - 2b^T Ax + b^T b\end{aligned}$$

- Taking gradient wrt  $x$ , we get

$$\nabla_x (x^T A^T Ax - 2b^T Ax + b^T b) = 2A^T Ax - 2A^T b$$

- Solving for  $x$  yields,  $x = (A^T A)^{-1} A^T b$

# Gradient of Determinant

- Consider  $A \in \mathbb{R}^{n \times n}$ , we want to find  $\nabla_A |A|$ , since

$$|A| = \sum_{i=1}^n (-1)^{i+j} A_{ij} |A_{\setminus i, \setminus j}|$$

so

$$\frac{\partial}{\partial A_{kl}} |A| = \frac{\partial}{\partial A_{kl}} \sum_{i=1}^n (-1)^{i+j} A_{ij} |A_{\setminus i, \setminus j}| = (-1)^{k+l} |A_{\setminus k, \setminus l}| = (\text{adj}(A))_{lk}$$

- $\nabla_A |A| = (\text{adj}(A))^T = |A| A^{-T}$

- Consider a function  $f : \mathbb{S}_{++}^n \mapsto \mathbb{R}, f(A) = \log |A|$
- $$\frac{\partial \log |A|}{\partial A_{ij}} = \frac{\partial \log |A|}{\partial |A|} \frac{\partial |A|}{\partial A_{ij}} = \frac{1}{|A|} \frac{\partial |A|}{\partial A_{ij}}$$
- Now it is obvious that

$$\nabla_A \log |A| = \frac{1}{|A|} \nabla_A |A| = A^{-1}$$